DO NOW — RECALL

Lesson 127 · Lossless and Lossy Compression

Name: _____ Class: _____ Date: _____

1. What is indirect addressing in a CPU instruction?

2. What is the binary result of 1 + 1 in a full adder with carry-in 0?

3. What is O(n) search complexity associated with?

4. Which memory is typically volatile: RAM or ROM?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The instruction points to a location holding the effective address.

2. Model answer

Sum 0, carry-out 1.

3. Model answer

Linear/sequential search in the worst case.

4. Model answer

RAM.

By the end, you can…

• Repeated patterns and predictable structures can be encoded efficiently.

• Lossless retains all detail; lossy permits irreversible approximation.

• Consider file size, compression ratio, quality and purpose.

Vocabulary — know these terms

lossless — Original data can be reconstructed exactly.

lossy — Some original information is deliberately discarded.

redundancy — Repeated or predictable data.

compression ratio — Original size divided by compressed size (by this convention).

NEW LEARNING

Read, annotate and explain

Find redundant data

Repeated patterns and predictable structures can be encoded efficiently.

Choose method

Lossless retains all detail; lossy permits irreversible approximation.

Measure impact

Consider file size, compression ratio, quality and purpose.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A medical image must remain identical after decompression. Choose a technique and justify.

Medical image data

Predict the first step from the prompt.

Lossless encoding

Explain the intermediate result or decision.

Exact reconstruction

Lossless compression is required if exact original image data must be preserved. Lossy compression permanently discards information and could alter diagnostically important features.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define lossy and lossless compression and give one application of each. [6]

Question 2

A file is reduced from 12 MB to 3 MB. Calculate the original-to-compressed compression ratio. [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate a compression choice for source code, an online photograph and an MRI scan. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A lossy method always reconstructs the original bit-for-bit.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.